## Configuring the Apache server

### Enable KeepAlive

Enabling KeepAlive in **httpd** allows persistent connections. These long-lived HTTP sessions allow multiple requests to be sent over the same TCP connection, as it does not require separate connection set-ups for each file. This reduces some overhead and significantly reduces latency periods. By default, Fedora's Apache **httpd** package has KeepAlive disabled. They should be enabled, with a timeout of two seconds. Don't keep this very high since it may overload your server. Take a look at Figure 1 to see the changes required in the Apache configuration file.

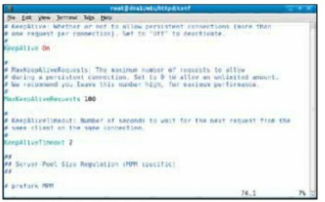



Figure 1: Enabling KeepAlive in Apache

### Handling of metadata

Metadata are typically defined as 'data about data'. When you try to install a package or update a system, the first things that get downloaded are package metadata. These are files with information about the packages, their age and other details. If, for example, a computer has old metadata cached, according to which all the packages are up-to-date, no new updates will be installed into the system. To work around this, we explicitly add the **Cache Control: must-revalidate** option, which insists that Yum or any client must revalidate the metadata against the server before serving it from the cache. For this, add the following section to your **/etc/httpd/conf/httpd.conf** around the **<Location>** directive (around line 900; take a look at Figure 2 to get an understanding of the exact location):




Figure 2: Configuring metadata handling in Apache

```
<LocationMatch "\.(xml|xml\.gz|xml\.asc|sqlite)">
  Header set Cache-Control "must-revalidate"
  ExpiresActive On
  ExpiresDefault "now"
</LocationMatch>
```

### Content types

ISO and RPM files should be served using **MIME Content-Type: application/octet-stream**. In Apache, this can be done inside a VirtualHost or similar section:

```
<VirtualHost *:80>
AddType application/octet-stream .iso
AddType application/octet-stream .rpm
</VirtualHost>
```

### Limiting download accelerators

Download accelerators will try to open the same file many times, and request chunks, hoping to download them in parallel. This can overload already heavily-loaded mirror servers, and cause a denial of service. In order to limit connections to ISO directories by some amount, per IP, add this to your apache configuration file:

```
<IfModule mod_limitipconn.c>
MaxConnPerIP 3
</IfModule>
```

To block ranged requests as this is, indeed, what download accelerators do, add this section to your apache configuration file:

```
RewriteEngine on
RewriteCond %{HTTP:Range} [0-9]$
RewriteRule \.iso$ / [R,L]
```

### Restart Apache

Now restart Apache. If everything is fine, you should not get an error. If you can start the Apache server successfully, it means you are done with most things.

## Registering your mirror

Now that your configuration is almost done, you must register your mirror, regardless if it is a private or a public mirror.

### Get a Fedora account

First, you need to go to **https://admin.fedoraproject.org/accounts/** and get yourself a Fedora account. Without this, you can't proceed further. However, you may choose to not sign the Fedora Contributor License Agreement (CLA), which is not required if you only want to be a mirror administrator. Still, do consider signing the CLA, as other contributions to Fedora require it, and with all the fun you're having as a mirror administrator, you may find other aspects of Fedora you want to be part of, as well.